

YELLOWSTONE REGIONAL, WY, USA

WMO: 726700

Lat: 44.517N

Lon: 109.017W

Elev: 1552

StdP: 84.02

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24045

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.7	-28.7	0.3	-20.7	-24.8	0.5	-18.6	12.3	2.8	11.4	3.1	2.0	30	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	32.5	15.2	31.1	14.8	29.0	14.4	16.8	27.9	16.0	27.3	15.3	26.5	4.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	11.2	20.4	12.1	10.7	20.2	11.0	9.9	19.8	53.1	27.3	50.6	27.4	48.4	26.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.0	8.1	-26.9	35.3	3.2	1.5	-29.2	36.4	-31.1	37.2	-32.8	38.0	-35.1	39.1
			-27.2	18.3	2.8	0.9	-29.2	19.0	-30.9	19.5	-32.4	20.0	-34.5	20.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-3.2	-2.5	2.7	6.1	11.2	16.5	21.6	20.3	15.0	7.7	1.5	-3.0
	DBStd	10.34	7.13	7.26	6.32	5.23	4.77	4.17	3.23	3.48	5.13	5.75	6.93	6.75
	HDD10.0	1967	408	351	234	137	43	4	0	0	15	110	262	404
	HDD18.3	4059	666	584	486	366	224	82	10	21	123	330	504	662
	CDD10.0	1198	0	1	7	21	81	200	358	320	164	39	7	0
	CDD18.3	247	0	0	0	0	3	27	110	83	23	1	0	0
	CDH23.3	2881	0	0	0	4	59	354	1245	927	277	14	0	0
	CDH26.7	1030	0	0	0	0	8	107	500	338	76	1	0	0
Wind	WSAvg	3.4	3.5	3.6	3.8	3.8	3.4	3.3	3.1	3.0	3.0	3.3	3.5	3.5
Precipitation	PrecAvg	232	9	6	10	23	46	41	24	21	26	17	10	7
	PrecMax	314	49	30	27	58	120	89	83	66	85	70	21	30
	PrecMin	148	0	0	0	2	2	2	5	1	0	1	0	0
	PrecStd	49	13	6	6	15	28	24	18	16	21	16	6	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.9	19.7	23.6	27.6	32.5	35.2	33.9	31.3	25.3	18.2	12.5
		MCWB	4.8	5.0	7.7	10.0	12.7	14.8	15.7	15.0	14.5	11.1	7.4	4.5
	2%	DB	10.3	12.2	17.1	20.9	25.0	29.7	32.8	32.1	28.8	22.2	16.1	10.8
		MCWB	3.3	4.0	6.2	8.4	11.9	14.1	15.5	14.7	13.7	9.7	6.6	3.9
	5%	DB	8.0	9.4	14.1	17.7	22.6	27.4	31.4	30.2	27.1	19.0	13.4	8.0
		MCWB	1.9	3.0	5.1	7.5	10.9	14.1	15.3	14.2	12.7	8.7	5.2	2.3
	10%	DB	7.1	7.4	12.3	15.5	20.3	25.1	29.2	28.4	24.0	17.0	11.7	6.5
		MCWB	1.3	1.4	4.4	6.6	10.3	13.4	14.9	14.0	11.7	7.7	4.5	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	5.8	8.0	10.6	14.4	17.3	18.1	17.1	16.2	12.0	8.7	5.3
		MCDB	11.4	13.1	18.3	22.0	23.7	27.1	28.3	27.6	26.3	22.6	16.8	11.2
	2%	WB	3.7	4.4	6.7	9.1	12.9	16.0	17.1	16.1	14.7	10.4	6.9	4.0
		MCDB	9.5	11.3	15.9	18.7	22.5	25.9	28.6	27.4	25.8	20.3	15.1	9.5
	5%	WB	2.4	3.1	5.6	8.0	11.8	15.0	16.4	15.4	13.5	9.2	5.6	2.6
		MCDB	8.0	9.1	13.8	16.9	21.1	24.9	28.1	26.7	24.5	17.9	12.9	8.0
	10%	WB	1.3	1.7	4.5	6.8	10.7	13.9	15.7	14.8	12.4	8.1	4.4	1.1
		MCDB	6.6	7.1	11.7	14.7	19.1	23.7	27.2	26.0	22.9	15.9	11.0	6.3
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.8	11.6	11.5	11.7	13.0	14.3	14.0	13.0	11.5	10.5	10.3
		MCDBR	11.8	12.6	14.0	15.2	15.6	16.2	16.2	16.5	16.0	15.0	12.2	11.2
		MCWBR	7.3	7.6	7.5	7.5	6.9	7.0	6.3	6.4	6.9	7.4	6.8	7.0
	5% WB	MCDBR	11.7	12.3	13.5	14.2	14.0	14.7	14.6	14.3	14.5	14.0	11.9	11.0
		MCWBR	7.6	7.7	7.4	7.3	6.6	6.9	6.2	6.1	6.6	7.4	6.9	7.3
Clear-Sky Solar Irradiance	taub	0.235	0.242	0.258	0.294	0.318	0.323	0.340	0.338	0.306	0.273	0.247	0.232	
	taud	2.512	2.518	2.519	2.440	2.447	2.471	2.453	2.435	2.525	2.585	2.565	2.519	
	Ebn at Noon	923	974	993	972	951	943	922	914	924	922	900	893	
	Edh at Noon	65	78	90	107	110	108	108	106	89	73	61	58	
All-Sky Solar Radiation	RadAvg	1.80	2.79	4.13	5.28	5.89	6.90	7.03	5.93	4.73	3.03	1.93	1.43	
	RadStd	0.14	0.20	0.32	0.39	0.55	0.50	0.35	0.27	0.36	0.24	0.17	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page